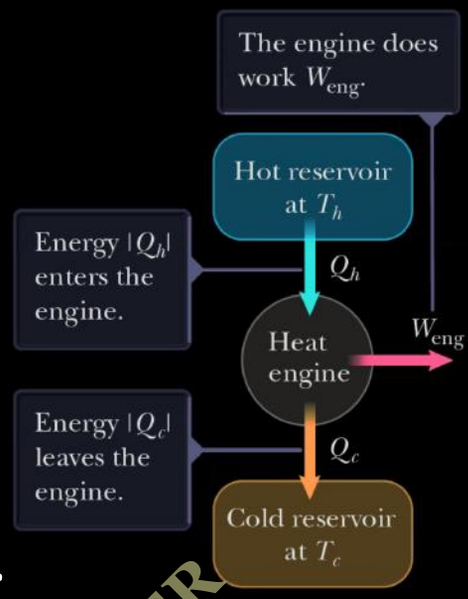
HEAT ENGINE: An engine is a device which absorbs energy from a reservoir at a high temperature and converts a portion of it in to work while rejecting a portion to a cold reservoir. The schematic of a simple engine is as shown in the adjacent figure. Typically any engine will have a working fluid which will be undergoing various thermodynamic processes in order to achieve the objective of the engine. (In practice, the engine might absorb heat not just at a single temperature, but at multiple temperatures).



EFFICIENCY OF AN ENGINE: Thermal efficiency of an engine is defined as ratio of net work done by the engine during one cycle to the energy input at higher temperature during the cycle. Mathematically:

$$\eta = \frac{W_{out}}{|Q_h|} = \frac{|Q_h| - |Q_c|}{|Q_h|} = 1 - \left|\frac{Q_c}{Q_h}\right|$$

Here Q_c is the heat rejected by the working fluid to the colder reservoir (or atmosphere). Since in one complete cycle, $\Delta U = 0 \therefore \Delta W = |Q_h| - |Q_c|$.

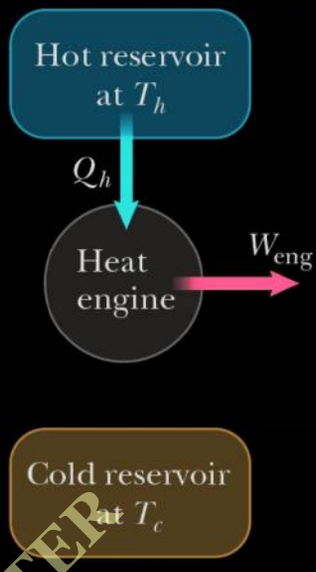
KELVIN PLANCK STATEMENT OF SECOND LAW: It is impossible to construct an engine working in a cycle which produces no other effect than the input of energy by heat from a reservoir and doing an equal amount of work.

It is a consequence of a more general formulation of second law in terms of entropy, which roughly says that thermodynamic systems spontaneously go from a statistically less probable state to a statistically more probable state. If a perfect engine were possible, it would be spontaneously creating "ordered energy" or work (think coordinated movement of piston particles) from a disordered

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energy. Think of random motion of particles in a furnace or a hot gas.

IMPOSSIBLE HEAT ENGINE →



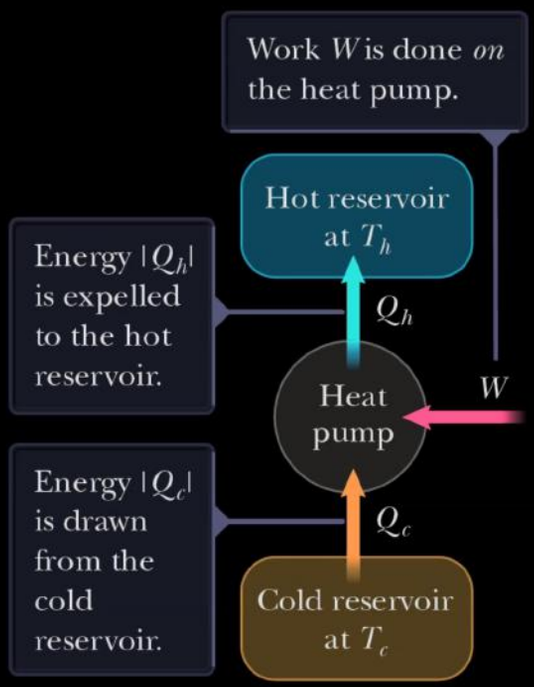
EXAMPLE: An engine transfers 2×10^3 J of energy from a hot reservoir during a cycle and exhausts 1.5×10^3 J to reservoir at cold temperature. (a) Find engine efficiency (b) What is the work output during one cycle (c) What is the power output if engine operates at 2000 rpm?

Solution: (a) $\eta = 1 - \frac{|Q_c|}{|Q_h|} = 1 - \frac{1.5\times10^3}{2\times10^3} = 25\%$ Ans.

(b) $\Delta W = Q_h - |Q_c| = (2-1.5)\times10^3 = 500$ Joule/cycle.

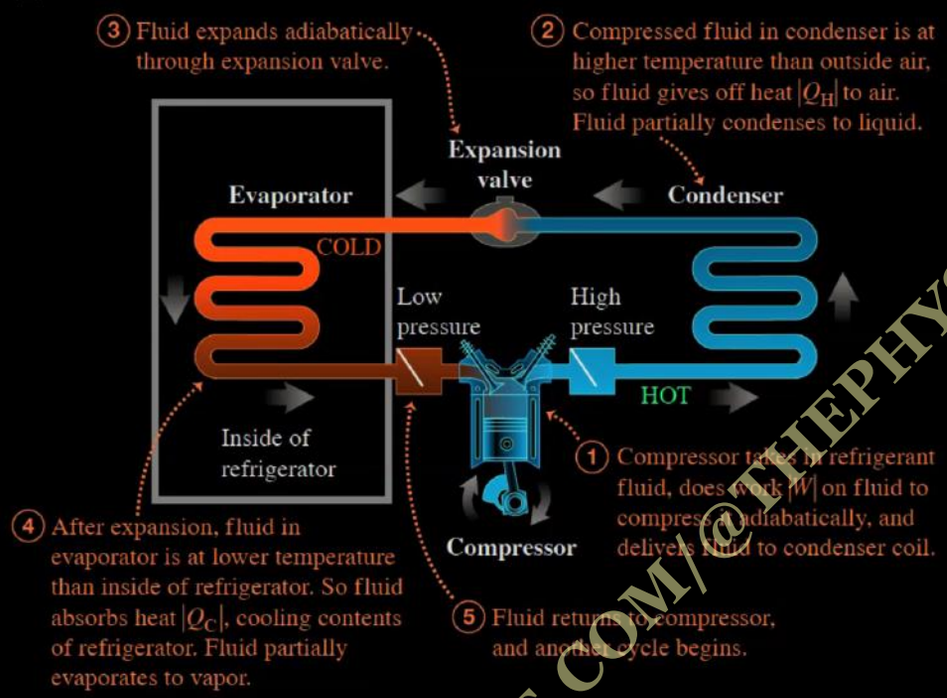
(c) $T = \frac{2\pi}{\omega}$, $P = \frac{\Delta W}{T} = \frac{\omega}{2\pi}\times\Delta W = \frac{2000\times2\pi}{2\pi\times60}\times500 \approx 1.7\times10^4$ W.

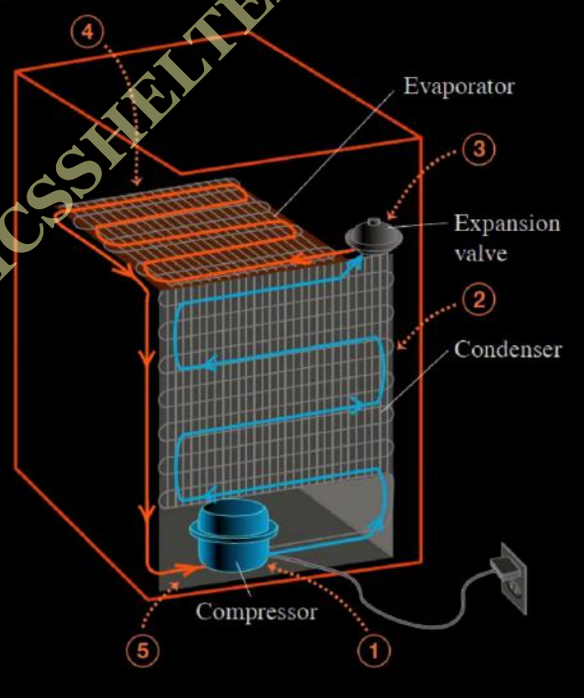
HEAT PUMPS & REFRIGERATORS: From energy and work flow point of view, a heat pump or a refrigerator is just the opposite of an engine. While in an engine, the purpose is to get some work by absorbing heat from a reservoir at high temperature and rejecting some heat to reservoir at cold temperature. In heat pump, the purpose is to pump heat from a reservoir at colder temperature and rejecting it to a reservoir at high temperature by doing some work on the system.



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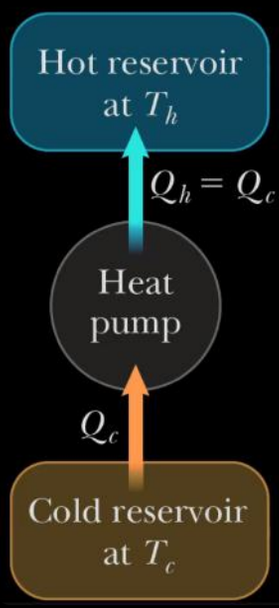
The same system may be called a refrigerator or heat pump depending on the purpose. If the purpose is to cool something then Q_L side is useful for us and the system is considered as refrigerator, however if the purpose is to heat something (Q_H side is useful) we consider the system as a heat pump. For instance a commonplace household refrigerator heats the room by extracting heat from vegetables and rejecting it to the room by utilising the compressor work. If the purpose here were to heat the room, we could consider the Refrigerator as the heat pump.





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CLAUSIUS STATEMENT OF SECOND LAW: It is impossible to construct a machine working in a cycle whose sole effect is to transfer energy continuously by heat from an object at low temperature and rejecting it to an object at higher temperature, without input of energy by work.



Impossible Heat Pump

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Coefficient Of Performance Of Refrigerator: The purpose of a refrigerator it to pump heat from colder temperature ($|Q_c|$) by expending work. For us it will be desirable to pump as much heat as possible by expending as little work as possible. Thus we define coefficient of performance of refrigerators as:

$$\text{COP (Cooling Mode)} = \frac{\text{Energy Transferred at low Temperature}}{\text{Work Done on Heat Pump}} = \frac{|Q_c|}{W}.$$

Coefficient Of Performance Of Heat Pump: For heat pump, the desirable part is the heat supplied at high temperature thus:

$$\text{COP (Heating Mode)}: \frac{\text{Energy Transferred at High Temp.}}{\text{Work Done on Heat Pump}} = \frac{|Q_H|}{W}$$

Example: A certain refrigerator has a COP of 5.00. When refrigerator is running power input is 500 W. A sample of water of mass 500 g at 20°C is placed in the freezer compartment. How long does it take to freeze the water to ice 0°C. Assuming other parts of refrigerator remain at same temperature.

Solution: $Q_L = mS\Delta T + mL = 500 \times 1 \times 20 + 500 \times 80 = 50{,}000$ cal.

Now $\text{COP} = \frac{Q_L}{W} = 5 \Rightarrow Q_L = 5 \times W = 5 \times P \times t$

$$\Rightarrow t = \frac{Q_L}{5 \times P} = \frac{50000 \text{ cal} \times 4.186 \text{ J/cal}}{5 \times 500 \text{ W}} = 83.7 \text{ Sec.}$$

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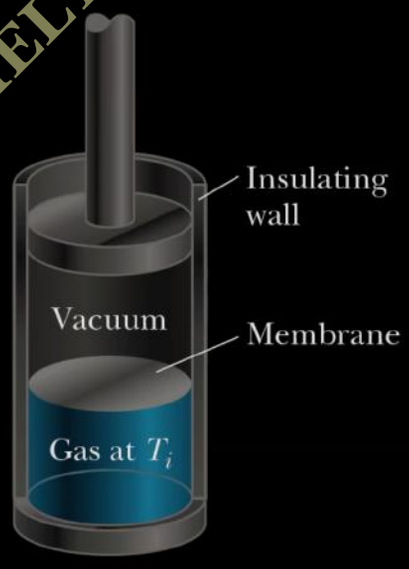
Reversible & Irreversible Processes:

In a reversible process, the system undergoing the process can be returned to its initial conditions along the same path on a PV diagram and every point along this path is an equilibrium state (Pressure and temperature of gas can be defined). For a process to be reversible:

(a) All heat transfer should occur across infinitesimal temperature difference.
(b) All piston movements should be infinitesimally slow (Across zero pressure diff).
(c) There should be no internal friction in the system.

Illustration: Why Free Expansion Is Irreversible:

Let us imagine a gas allowed to expand freely by puncturing the membrane as shown in adjacent figure. Now suppose we wish to restore the gas to initial conditions (volume as well as temperature). Then first we can compress the gas by pressing the piston. Here surroundings is expending "Organized Energy", some agency in the surrounding would be doing this work. Now upon compressing gas will get heated up and to restore its temperature, it should lose heat to the surrounding or release "Unorganized" energy. Thus once again we are making changes to the surroundings. The overall effect is that "organized energy" from surroundings is converted to "unorganized" energy. One might further imagine that Heat liberated from the gas could be used to power the engine providing work to compress the gas, however 100% of this heat can't be converted to work without rejecting some heat to reservoir at colder temperature (Kelvin Planck statement). Thus there is a net effect on surroundings and hence process is irreversible.



→ During free expansion, we can't talk about pressure of the gas until new equilibrium is established due to random changes occuring in the system. (Infact vacuum side pressure is zero). Thus irreversible process can't be represented on PV diagram.).

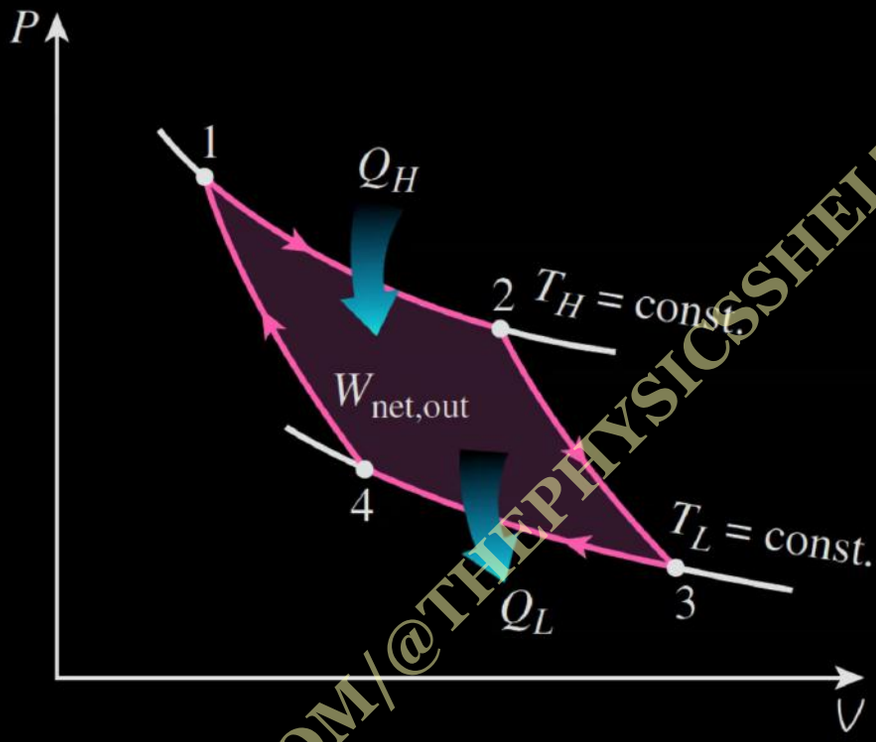
→ All practical processes are irreversible however if a process occurs very slowly,

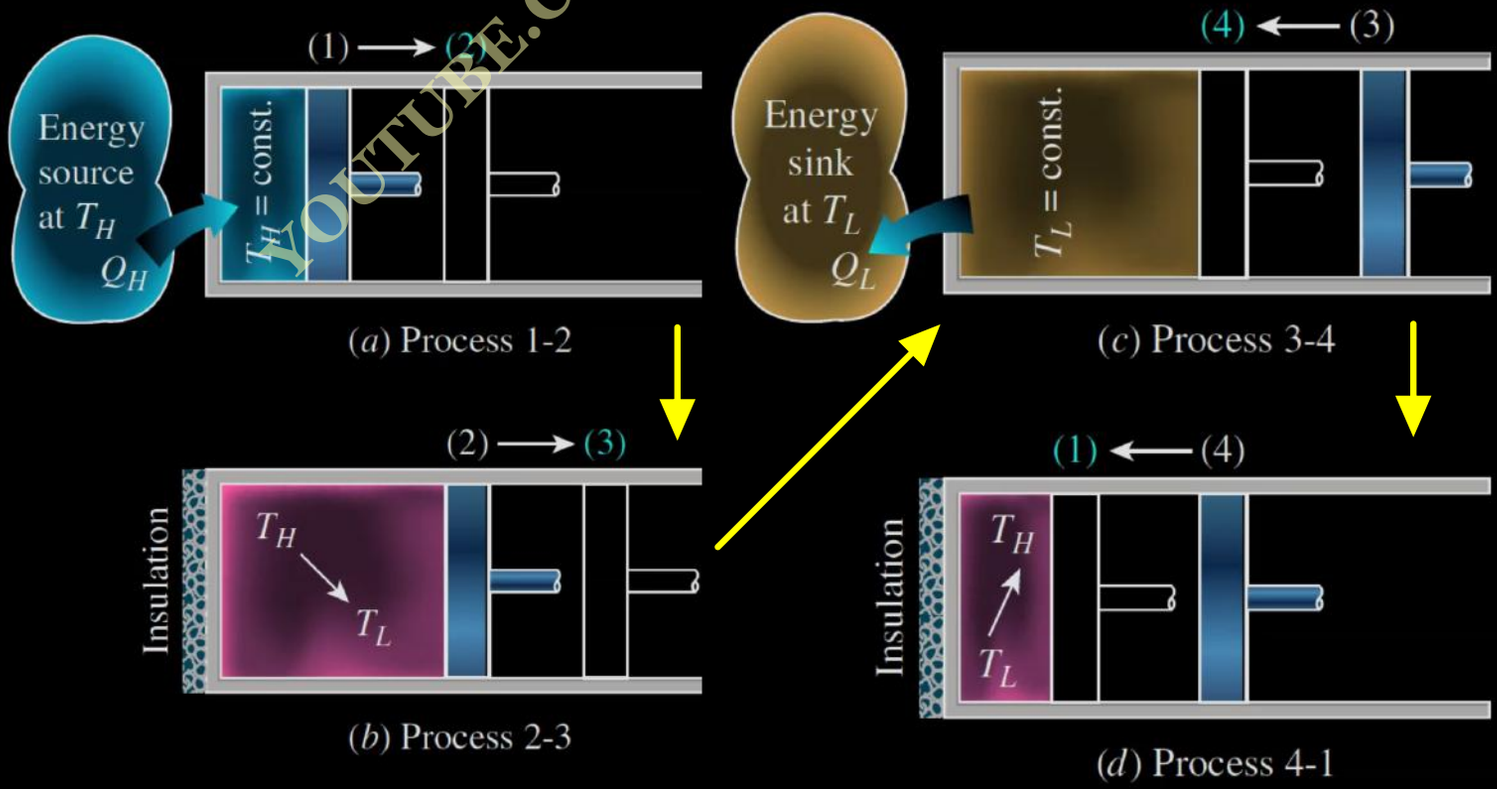
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we can say approximately that it is reversible

→ Internal friction converts mechanical energy (organized) in to internal thermal energy (unorganised). This effect can't be reversed completely and is therefore irreversible.

CARNOT ENGINE: It is a cyclic engine comprising of two isothermal curves and two adiabatic curves. The heat is input along the isothermal curve at high temperature and rejected along isotherm at lower temp.

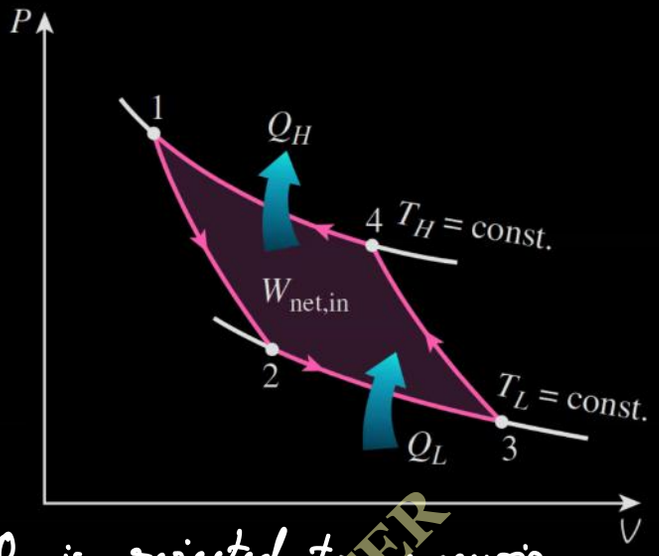




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Reversed Carnot Cycle (Carnot Refrigerator):

The Carnot cycle described in previous section is completely reversible thus all processes involved in it can be reversed in which case it becomes carnot refrigeration cycle. The cycle remains exactly same except that all heat and work interactions are reversed. Thus here a heat Q_L is absorbed from low temperature reservoir and Q_H is rejected to reservoir at higher temperature with help of a work input.

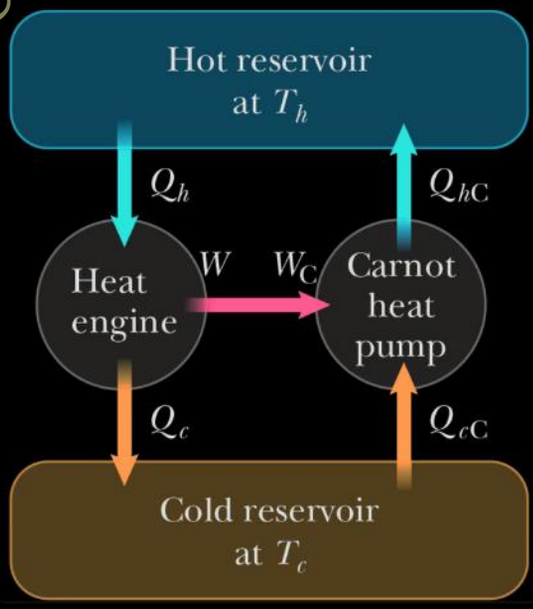


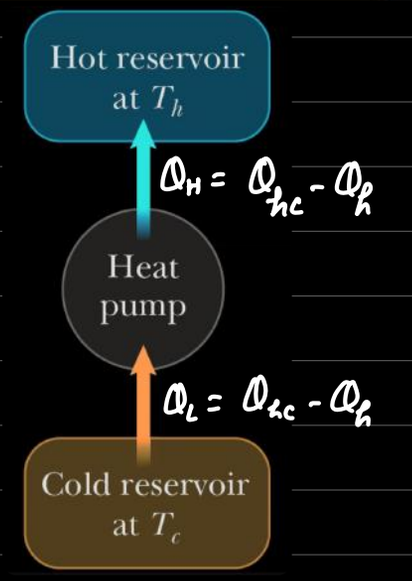
Carnot's Theorem: No Real (Irreversible) Engine operating between two reservoirs can be more efficient than a Carnot Engine operating between same two reservoirs.

We can prove this by showing that if such an engine were possible, we will be able to contradict the clausius statement of second law by clubbing such an engine with a reversed carnot engine (Carnot Heat pump).

Let us club such a superengine with a carnot heat pump which absorbs a work which exactly matches the work output of the superengine. Now since the super engine is more efficient than carnot engine, it absorbs less heat than a carnot engine to produce same amount of work. $\therefore$ $Q_h < Q_{hC}$. Thus the combined effect of the coupled device is to pump a heat $Q_{hC} - Q_h$ from a reservoir at lower temperature to a reservoir at higher temperature without any work input in violation of clausius statement.

$\Delta U = 0$ ($\because$ Cycle) $\Rightarrow \Delta Q_{in} = \Delta Q_{out}$



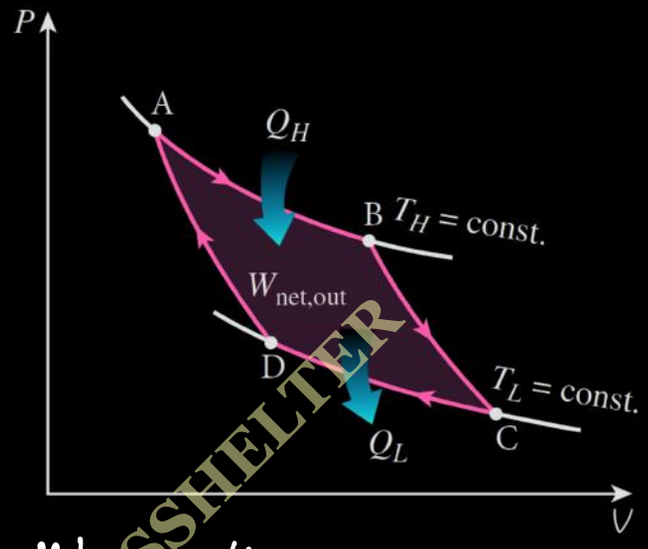


COMBINED EFFECT

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CARNOT'S SECOND PRINCIPLE : Any two reversible engines operating between same pair of reservoirs have same efficiency irrespective of working fluid.

VOLUME RATIOS IN CARNOT CYCLE:



Recall that for an adiabatic process, ideal gases follow $TV^{\gamma-1} = \text{Constant}$.

In our graph DA and BC are adiabats.

$$\Rightarrow \quad T_L V_D^{\gamma-1} = T_H V_A^{\gamma-1} \quad (1)$$

also:

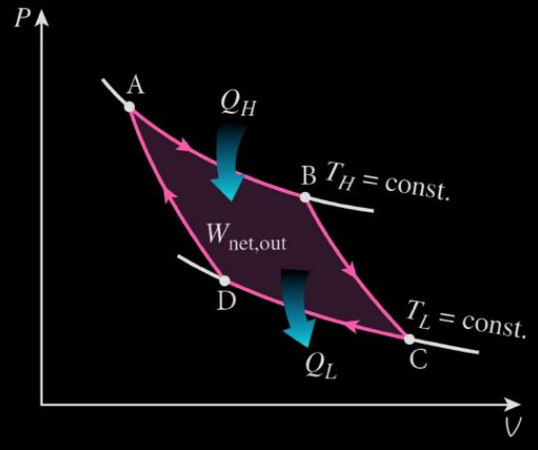
$$T_L V_C^{\gamma-1} = T_H V_B^{\gamma-1} \quad (2)$$

From (1) & (2):

$$\boxed{\frac{V_C}{V_D} = \frac{V_B}{V_A} = r}$$ $r \rightarrow$ Volume ratio.

EFFICIENCY OF CARNOT CYCLE:

$$\eta = 1 - \frac{|Q_L|}{|Q_H|}$$



Now For $A \rightarrow B$, $Q_H = \Delta W = nRT_H \ln\left(\frac{V_B}{V_A}\right)$

$$\Rightarrow |Q_H| = nRT_H \ln(r)$$

For $C \rightarrow D$, $Q_L = \Delta W = nRT_L \ln\left(\frac{V_D}{V_C}\right) = nRT_L \ln\left(\frac{1}{r}\right)$

$$\Rightarrow Q_L = -nRT_L \ln(r) \text{ or } |Q_L| = nRT_L \ln(r)$$

Now $$\eta = 1 - \frac{|Q_L|}{|Q_H|} = 1 - \frac{nRT_L \ln(r)}{nRT_H \ln(r)} = 1 - \frac{T_L}{T_H}$$

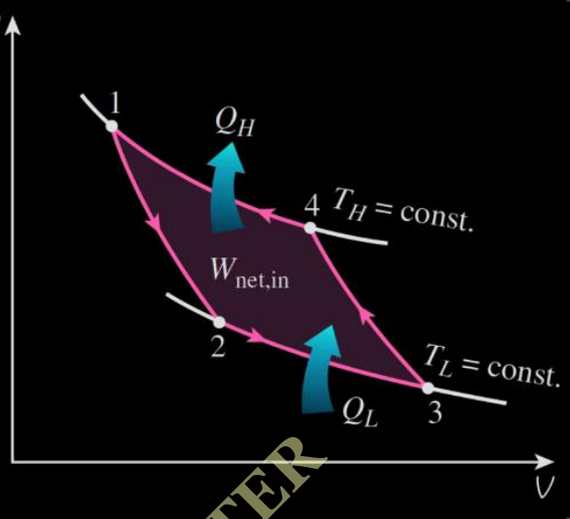
$$\Rightarrow \boxed{\eta_C = 1 - \frac{T_L}{T_H}} \rightarrow |Q_H| : |Q_L| = T_H : T_L \leftarrow \text{Note}$$

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Carnot Heat Pump (COP in cooling and Heating Modes):

$$COP(\text{Heating}) = Q_H/W = Q_H/(Q_H - Q_L)$$

$$\Rightarrow \boxed{COP_{HP} = \frac{T_H}{T_H - T_L} = \frac{1}{\eta_{CE}}}$$



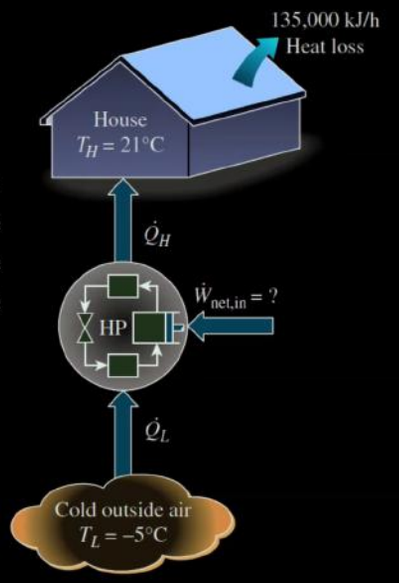
$$COP(\text{Cooling}) = Q_L/(Q_H - Q_L)$$

$$\Rightarrow \boxed{COP_R = \frac{T_L}{T_H - T_L} = COP_{HP} - 1 = \frac{1}{\eta_{CE}} - 1}$$

Example: (Cengel 6.7):

EXAMPLE 6–7 **Heating a House by a Carnot Heat Pump**

A heat pump is to be used to heat a house during the winter, as shown in Fig. 6–52. The house is to be maintained at 21°C at all times. The house is estimated to be losing heat at a rate of 135,000 kJ/h when the outside temperature drops to −5°C. Determine the minimum power required to drive this heat pump.



Solution: We need to replenish energy equal to the lost energy. $\therefore \dot{Q}_H = 135\,000$ KJ/hr.

Now $COP = \frac{T_H}{T_H - T_L} = \frac{\dot{Q}_H}{\dot{W}} \Rightarrow \dot{W} = \dot{Q}_H\left(\frac{T_H - T_L}{T_H}\right)$

$$\Rightarrow W = 135\times10^6\ J/hr \times \frac{26}{294} = 11.93\times10^6\ J/hr = 3.32\ kW \text{ ans.}$$

Note that this is the theoretical minimum required work, actual required work will be more than this (Carnot Heat pump has max. COP).

Efficiency of an Arbitrary cycle: So far we have considered the engines which exchange heat only at constant temperatures, however we can also have cycles where heat is exchanged at varying temperatures of working fluid. (Say isobaric or isochoric processes). In such cycles,

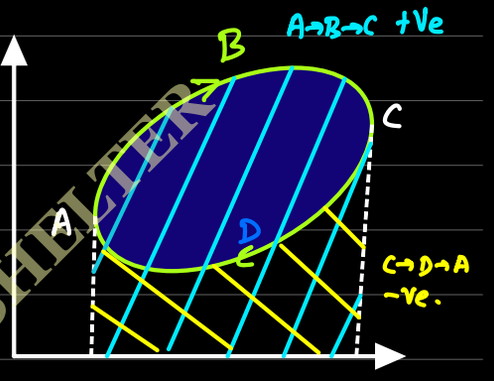
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while considering our energy expenditure we only count the heat which is positive (i.e supplied to the system). Thus for such cycles, efficiency will be given as:

$$\eta = \frac{\Sigma \Delta W}{\Sigma \Delta Q_+}$$

Difference between Engine cycle and Refrigeration Cycles. : Note that area under the PV diagram represents the work done in the cycle. Thus all engine cycles must be clockwise.

Refer to the adjacent figure. During A→B→C, the gas expands ∴ work done is +ve while during C→D→A, work done is negative (Gas is getting compressed). Thus net work done is positive and equal to area enclosed in the cycle ABCDA. (Since positive work (Turquoise Hatch) is more than negative work (Yellow Hatch) ∴ net work is positive. Had the cycle been anticlockwise, the work done would have been negative.

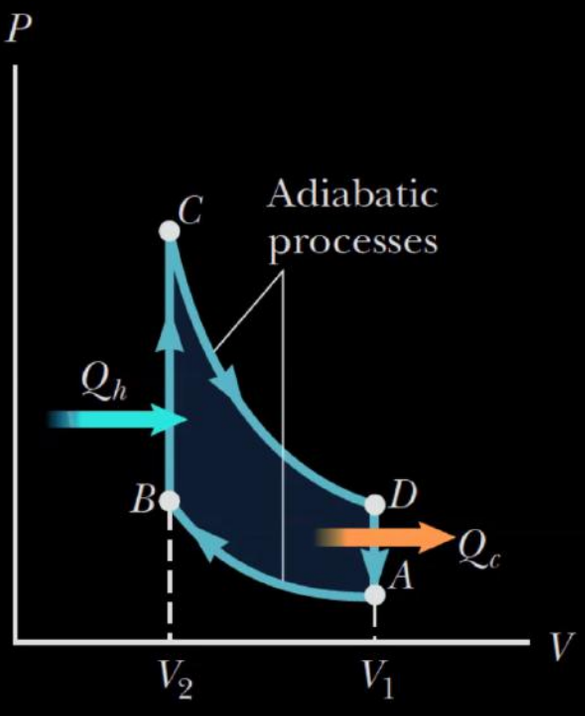


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Example: Otto Cycle

An otto cycle comprises of two adiabatic and two isochoric processes as shown in the figure. Prove that efficiency of the otto cycle is given as

$$\eta = 1 - \frac{1}{(V_1/V_2)^{\gamma-1}}$$



Solution: Recognise that in the cycle as a whole $\Delta U = 0$ ∴ $\Sigma \Delta Q = \Sigma \Delta W$.

Here the heat is supplied at constant volume V_2 and rejected at constant Volume V_1.

Thus $\Delta W = Q_h - |Q_c|$ (See figure).

and $\Delta Q_+ = Q_h$

$$\Rightarrow \eta = (Q_h - |Q_c|)/Q_h = 1 - \frac{|Q_c|}{Q_h}$$

Now: $P_C V_2^{\gamma} = P_D V_1^{\gamma}$ (1) $\Rightarrow \frac{P_C}{P_B} = \frac{P_D}{P_A} = r$ (3)

and $P_B V_2^{\gamma} = P_A V_1^{\gamma}$ (2)

Now $Q_H = \frac{f}{2} \Delta(PV) = \frac{f}{2} (P_C - P_B) V_2$, $|Q_C| = \frac{f}{2} (P_D - P_A) V_1$

$$\Rightarrow \eta = 1 - \left(\frac{P_D - P_A}{P_C - P_B}\right)\frac{V_1}{V_2} = 1 - \frac{P_D}{P_B}\left(\frac{P_D/P_A - 1}{P_C/P_B - 1}\right)\frac{V_1}{V_2} = 1 - \frac{P_A}{P_B}\left(\frac{r\gamma - 1}{r - 1}\right)\frac{V_1}{V_2}$$

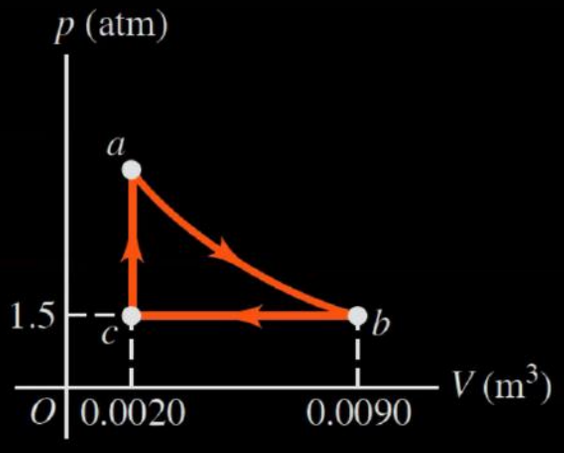
$$\Rightarrow \eta = 1 - \frac{P_A}{P_B}\cdot\frac{V_1}{V_2} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma - 1} \quad (\because (2))$$

$$\Rightarrow \boxed{\eta_{otto} = 1 - \frac{1}{(V_1/V_2)^{\gamma - 1}}}$$

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EXAMPLE: YOUNG & FREEDMAN:

20.5 •• The pV-diagram in **Fig. E20.5** shows a cycle of a heat engine that uses 0.250 mol of an ideal gas with $\gamma = 1.40$. Process ab is adiabatic. (a) Find the pressure of the gas at point a. (b) How much heat enters this gas per cycle, and where does it happen? (c) How much heat leaves this gas in a cycle, and where does it occur? (d) How much work does this engine do in a cycle? (e) What is the thermal efficiency of the engine?



Solution: (a) For finding pressure at a, we will use the fact that it lies on isochore through c and adiabat through b.

$$p_a V_a^{\gamma} = p_b V_b^{\gamma} \Rightarrow p_a = \left(\frac{V_b}{V_a}\right)^{\gamma} \times p_b$$

$$\Rightarrow p_a = 1.5 \times 4.5^{1.4} \text{ atm} = 12.3 \text{ atm.}$$

(b) Here the heat enters the gas only along $c \to a$ [Isochoric heating]. $\Delta Q_H = \Delta U + \Delta W = \Delta U + 0 = \Delta U$

$$\Rightarrow \Delta Q_H = \Delta U = \frac{f}{2}\Delta(PV) = \frac{\Delta(PV)}{\gamma - 1} = \frac{(12.3 - 1.5) \times 0.002 \text{ atm m}^3}{0.4}$$

$$\Rightarrow \Delta Q_H = 5.4 \times 10^{-2} \text{ atm m}^3$$

(c) Heat is rejected along $b \to c$ (Isobaric cooling).

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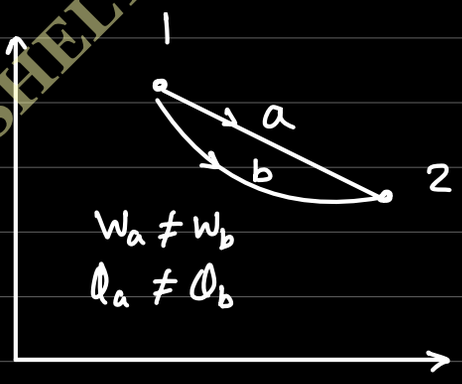
$$|\Delta Q| = n C_p |\Delta T| = \frac{\gamma R}{\gamma - 1} n \Delta T = \left| \frac{\gamma}{\gamma - 1} \Delta (PV) \right| = \frac{\gamma}{\gamma - 1} |P(\Delta V)|$$

$$|\Delta Q_c| = \frac{1.4}{0.4} \times 1.5 \, (0.009 - 0.002) = 3.675 \times 10^{-2} \text{ atm } m^3$$

(d) Now $\eta = 1 - \frac{|Q_c|}{Q_H} = 1 - \frac{3.675}{5.4} \simeq 0.319 = 31.9\%$ Ans.

State Property Vs Path Property: A quantity whose value depends on the path followed on a thermodynamic graph (Say PV graph) is known as path property (e.g. Work done in going from one state to another. Or Heat supplied in going from one state to another.

Whereas a quantity whose value depends on the state of the thermodynamic system and not the path followed in achieving the state is called state property (e.g. U, T, P, f etc).

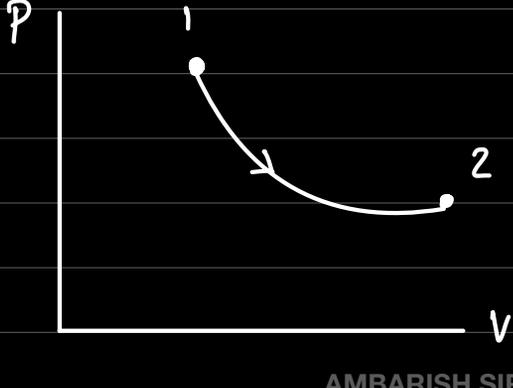


Thus changes in state properties only depend on initial and final states. If the change in certain quantity is independent of path then it is a sufficient condition to consider it as a state property.

Entropy: Suppose we define a quantity as $\Delta S = \int_1^2 \frac{dQ_r}{T}$, where dQ_r is the heat supplied to the system along any reversible elemental path and 1 and 2 are certain thermodynamic states then we can easily show for any ideal gas, this property is independent of path. The quantity S is called entropy. Just like internal energy, we are mostly interested in change of entropy rather than absolute entropy.

Proof That S is A State Property:

$$\Delta S = \int_1^2 \frac{dQ}{T} = \int_1^2 \frac{dU}{T} + \frac{dW}{T} = \int_1^2 \frac{nf}{2} R \frac{dT}{T} + \frac{P dV}{T}$$



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$$\Rightarrow \Delta S = \int_1^2 n\frac{f}{2}R\frac{dT}{T} + \int_1^2 \frac{nRT}{V}\cdot\frac{dV}{T} = n\frac{f}{2}R\ln\left(\frac{T_2}{T_1}\right) + nR\ln\left(\frac{V_2}{V_1}\right)$$

$$\Rightarrow \boxed{\Delta S = n\frac{f}{2}R\ln\left(\frac{T_2}{T_1}\right) + nR\ln\left(\frac{V_2}{V_1}\right)}$$

Thus we see that to find ΔS we did not need the equation of the path at all, T_1, T_2, V_1 and V_2 only depend on states and not how the transition between states occurs. Thus:

Entropy is a state property.

Note: During Irreversible process temperature of gas is not well defined and the process can't be represented on a PV graph however initial and final states can always be represented on PV diagram and for entropy change calculation this is sufficient thus:

→ Even if the change is happening irreversibly, entropy change is always calculated by imagining a reversible process between initial and final state.

EXAMPLE: YOUNG & FREEDMAN.

EXAMPLE 20.6 Entropy change in a temperature change

One kilogram of water at 0°C is heated to 100°C. Compute its change in entropy. Assume that the specific heat of water is constant at 4190 J/kg · K over this temperature range.

Solution: $dS = \frac{dQ_{rev}}{T} = \frac{mSdT}{T} \Rightarrow \Delta S = mS\int_{T_1}^{T_2}\frac{dT}{T} = mS\ln\frac{T_2}{T_1}$

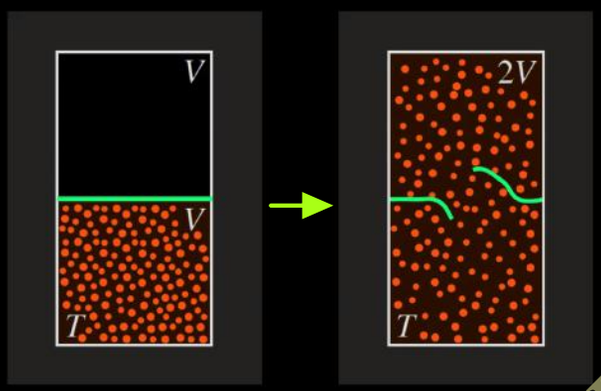
$\Rightarrow \Delta S = 1\text{kg} \times 4190\,\text{J/kgK} \times \ln(373/273) = 1.31\times10^3\,\text{J/K}$ Ans.

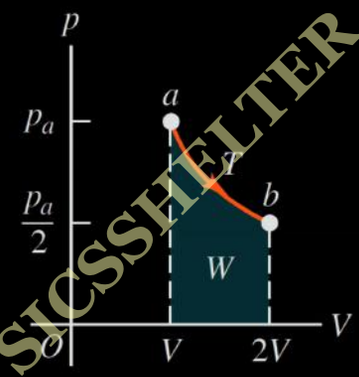
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Example: Young & Freedman:

EXAMPLE 20.8 Entropy change in a free expansion

A partition divides a thermally insulated box into two compartments, each of volume V (**Fig. 20.18**). Initially, one compartment contains n moles of an ideal gas at temperature T, and the other compartment is evacuated. We break the partition and the gas expands, filling both compartments. What is the entropy change in this free-expansion process?





Imagined Reversible Procss.

Wrong Solution: $\Delta S = \frac{\Delta Q}{T}$ $\because \Delta Q = 0$ $\therefore \Delta S = 0$ ✗ WRONG.

Solution: Note that entropy change is always defined as $\Delta S = \int \frac{dQ_{rev}}{T}$ and not $\int \frac{SQ}{T}$, $\because$ even if the process is irreversible

We have to imagine a reversible process between initial and final configuration. Now recall that for any ideal gas,

$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) + n\frac{f}{2}R \ln\left(\frac{T_2}{T_1}\right)$. For our case $T_2 = T_1$ $V_2 = 2V_1$

$\Rightarrow \quad \Delta S = nR \ln(2)$ Ans.

Example: Young & Freedman:

EXAMPLE 20.10 Entropy change in an irreversible process

Suppose 1.00 kg of water at 100°C is placed in thermal contact with 1.00 kg of water at 0°C. What is the total change in entropy? Assume that the specific heat of water is constant at 4190 J/kg · K over this temperature range.

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Sol: Even if the process is irreversible, we have to imagine a reversible process that leads to the same final state. Here since temperature difference is finite, ∴ there is an irreversible heat flow. However we can always add & subtract heat dQ by ∂Q s.t that at every step the temperature is well defined. Here final temperature will be 50°C (∵ masses are equal ∴ temperature is simply the average). Now we can consider the final state to be a result of two reversible processes.
① Reversible heating of cold water ② Reversible cooling of hot water.

For Hot water, $dS_H = \frac{dQ}{T} = ms\frac{dT}{T} \Rightarrow \Delta S_H = ms\ln\left(\frac{T_f}{T_i}\right)$

$\Rightarrow \Delta S_H = 1Kg \times 4190 J/KgK \times \ln\left(\frac{323}{373}\right) = -603\ J/K$

Similarly for cold water $\Delta S_c = 1Kg \times 4190 \times \ln\left(\frac{323}{273}\right) = +705\ J/K$

Thus net change in entropy $= \Delta S_H + \Delta S_c = -603 + 705 = +102\ J/K$.

Entropy Change In A Reversible Adiabatic Process: In any reversible Adiabatic process dQ is always zero. ∴ $dS = \frac{dQ}{T} = 0 \Rightarrow$

$\Delta S = \int \frac{Q}{T} dT = 0$. Thus entropy change is zero.

∴ **A Reversible Adiabatic Process Is Also "Isentropic" Process.**

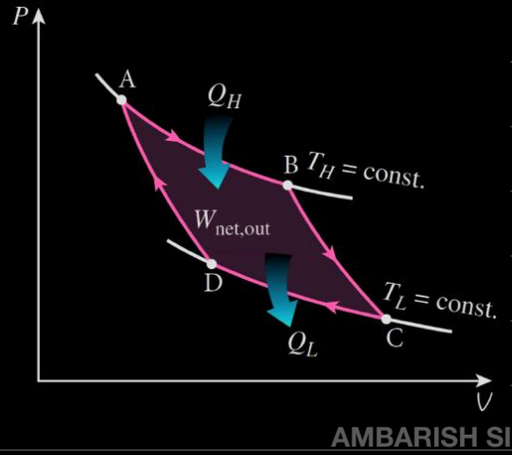
Entropy Change In A Carnot Cycle:

Consider the adjacent carnot cycle,

$\Delta S_{AB} = \frac{Q_H}{T_H}$, $\Delta S_{BC} = 0$, $\Delta S_{CD} = -\frac{Q_L}{T_L}$

$\Delta S_{DA} = 0$ ∴ $\Delta S_{cycle} = \frac{Q_H}{T_H} - \frac{Q_L}{T_L}$

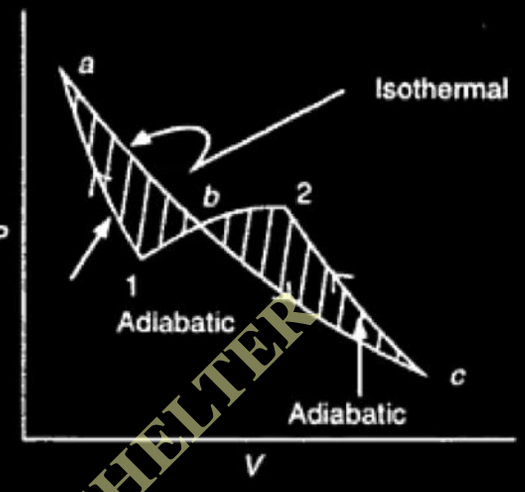
But $Q_H/T_H = Q_L/T_L = 0$ ∴ $\Delta S_{cycle} = 0$



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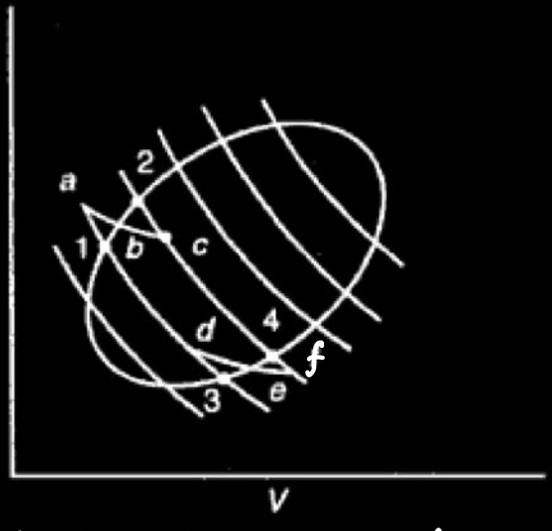
Here we have proved that entropy change in any Carnot cycle is always zero.

Approximating Arbitrary Process By Isotherms & Adiabats :



Consider segment of an arbitrary process $1 \to b \to 2$. It can be easily seen that this segment can be easily approximated by a combination of isotherms and adiabats. (In the sense that we can get same amount of work and internal energy change by drawing an alternative process between 1 & 2 comprised of Adiabats and isotherms.). Here if we consider process $1-a-b-c \to 2$, such that $1 \to a$ and $c \to 2$ are adiabats and $a \to c$ is an isotherm, we can always choose the points a and c such that Area $1 \to a \to b-1$ is equal to area $bc2b$, in that case net work done in cyclic process $1\,a\,b\,c\,2\,b\,1$ will be Zero because it can be seen as a combination of a clockwise cycle $(1\,a\,b\,1)$ and an anticlockwise cycle $b\,c\,2\,b$. ∴ we can say that work done along $1 \to b \to 2$ must be same as that in process $1 \to a \to b \to c \to 2$.

Approximating Arbitrary Cycle By Carnot Cycles:



Based on discussion in previous section, segment segment $1\,b\,2$ is thermodynamically identical to $1\,a\,b\,c\,2$ and segment $4\,e\,3$ is identical to segment $4\,f\,e\,d\,3$. Thus sum of entropy changes along segments $1 \to 2$ and $4 \to 3$ of the original cycle is same as that for carnot cycle $a \to c \to f \to d \to a$ (Note that entropy change for adiabats is zero). Extending this argument for other segments of the cycle we can see that in this

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manner, the original cycle can be thought of as a combination of several infinitesimal carnot cycles and since entropy change in any Carnot cycle is zero ∴ Entropy change in any arbitrary cycle is also zero. If change in any property in a complete cycle is identically zero then it is a sufficient condition for the process to be a state Property. ∴

→ ENTROPY MUST BE A STATE PROPERTY.

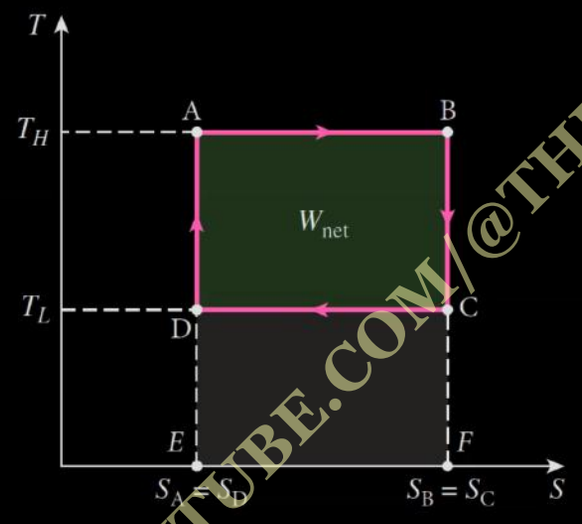
→ ENTROPY CHANGE IN EVERY CYCLIC PROCESS IS ZERO.

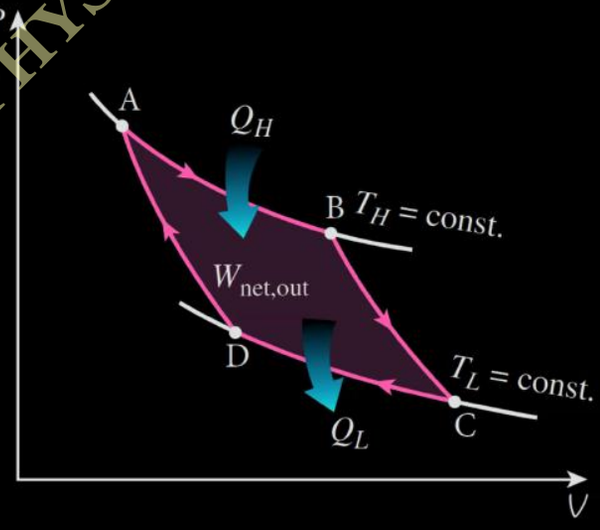
EXAMPLE: (CENGEL):

EXAMPLE 7–6 The *T-S* Diagram of the Carnot Cycle

Show the Carnot cycle on a *T-S* diagram and indicate the areas that represent the heat supplied Q_H, heat rejected Q_L, and the net work output $W_{net,out}$ on this diagram.

Solution:





Recall that $dS = \frac{dQ}{T} \Rightarrow dQ = TdS$ ∴ Area under the T-S diagram represents the heat supplied in any process. Also recall that adiabatic processes are also isentropic. This leads to the above T-S diagram of the Carnot cycle. Now:

Q_H = Area under A→B i.e Area ABFEA rectangle.

Similarly, (Q_L) = Area of rectangle DCFED. (Q_L is negative)

Now Work done in the cycle : $\Delta W = Q_H - |Q_L|$

Thus work done is the area enclosed by the cycle i.e. ABCDA.

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Statistical Interpretation Of Entropy:

Macrostate: A macrostate is a constraint on a system of particles, for example for a thermodynamic system under a piston, the macrostate could be the constraint on total number of particles, total energy and total Volume occupied by the system.

Microstate: The exact possible state of a system under the given macroscopic constraints is known as microstate. For instance, a microstate would comprise of particular velocity and space coordinates for a point particle gas system.

To properly understand microstates and macrostates, we can consider a system of four coins, where each coin can be flipped in to a "Head" or "Tail" configuration. Subject to the macroscopic constraint, following microstates are possible:



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Even in such a small sample it can be seen that the microstates with two heads and two tails are much more probable than the state with all heads or all tails. This effect becomes much more pronounced as the number of particles increases.

Similarly in thermodynamic systems, the states where all the energy is distributed to very few particles is exponentially negligible in comparison to the states where energy is distributed among all the particles. In general, upon a random observation,

any thermodynamic system will be found in a type of microstate which has statistically much larger number of ways of producing. Each microstate is equally probable, but there are exponentially larger ways of producing certain types of microstates. For instance consider a room divided into a cubical grid in which you wish to distribute gas particles. In this situation the number of ways to distribute all particles in left half of the room would be exponentially lower than the number of ways of distributing such that both left and right half of the room have nearly equal number of particles. Thus upon a random observation it will be found that both halves of the room have nearly equal number of particles. [Microstates keep switching randomly and therefore systems are more likely to be in states which have more number of ways of generation).

STATISTICAL DEFINITION OF ENTROPY: For any macroscopic state of the system, if Ω is the number of possible microstates then the entropy of the state is given as:

$$\boxed{S = k \ln \Omega}$$ $k \rightarrow$ Boltzmann Constant.

Which also means: $\Delta S = k \ln \Omega_f - k \ln \Omega_i = k \ln\left(\frac{\Omega_f}{\Omega_i}\right)$

Where Ω_f and Ω_i are final fugacities (no of microstates).

$$\boxed{\Delta S = k \ln(\Omega_f/\Omega_i)}$$

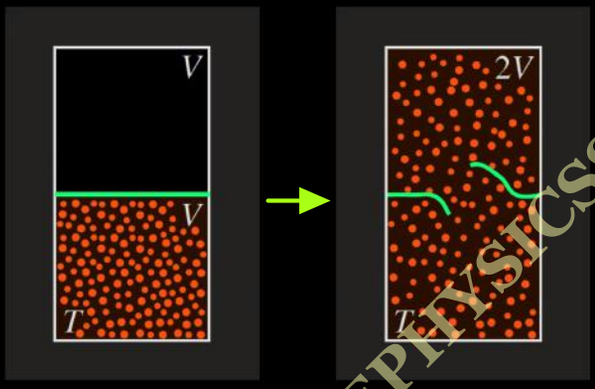
Here we shall not rigorously prove equivalence of thermodynamic and statistical entropy, However recall that for isothermal process the thermodynamic entropy change was given as

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) \quad \left(\because \Delta S = nR \ln \frac{V_2}{V_1} + \frac{nfR}{2} \ln(T_2/T_1)\right).$$

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It can be seen that number of "Place Holder" in a room would increase in proportion to the Volume, ∴ the ratio Ω_f/Ω_i must be proportional to (V_2/V_1). Similarly if the process were isochonic, higher temperature would mean greater energy available for distribution and therefore (Ω_f/Ω_i) would be proportional to (T_2/T_1). We shall get a little more insight in to it from a following example.

EXAMPLE: Calculate the entropy change in free expansion of n moles of a gas to double its occupied volume.



Solution: We shall do an abstract calculation keeping in mind that the number of particles is very large. Let V_m be volume of each molecule (Think cubical molecules) and V be the volume of lower chamber. Then:

Total number of Available Place holders : $i = (V/V_m)$

Total number of particles : $N = n N_A$.

Now first particle we have i choices for placements
for second particle we have $i-1$ choices for placement, but since i is very large ∴ $i-1 \approx i$ or for second particle also we have almost i choices similarly for each particle.

Thus total permutations possible for spatial arrangements:

$$\Omega_i = i \times i \times \dots \quad (N \text{ terms})$$

$$\Rightarrow \Omega_i = i^N$$

Now when volume doubles, number of available placeholders :

$$2i = (2V/V_m).$$

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Now each particle has 2i number of choices for placements
∴ Final fugacity : $\Omega_f = (2i)^N$

Now change in entropy = $k \ln(\Omega_f/\Omega_i) = \Delta S$

$$\Rightarrow \quad \Delta S = k \ln((2i)^N / i^N) = k \ln 2^N = kN \ln(2)$$

$$\Rightarrow \quad \Delta S = k \times n \times N_A \ln(2) = nR \ln(2).$$

This is the same expression that we obtained earlier using thermodynamic definition of entropy.

→ Note that here we omitted energy permutations however it is ok here because total internal energy is same here and ∴ initial and final energy permutations are equal and makes no difference to ratio (Ω_f/Ω_i).

→ In Boltzmann's formulation of entropy, all microstates are considered equally probable however Gibbs had given a more comprehensive formulation which accounts for the fact that in certain thermodynamic systems all microstates might not be equally probable.

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CLAUSIUS INEQUALITY (SECOND LAW): For any thermodynamic cycle (reversible or irreversible), cyclic integral of $\delta Q/T$ is always less than or equal to zero, Mathematically:

$$\oint \frac{\delta Q}{T} \leq 0$$

This can be understood from our theory of cyclic engines.
Recall that for any reversible engine, $\oint \frac{dQ}{T} = 0$, now

suppose we have an irreversible engine between two reservoirs, then an irreversible engine will reject more heat and do less work as compared to a carnot engine. Thus ΔQ_2 (Heat rejected at lower temperature is more negative as compared to carnot engine

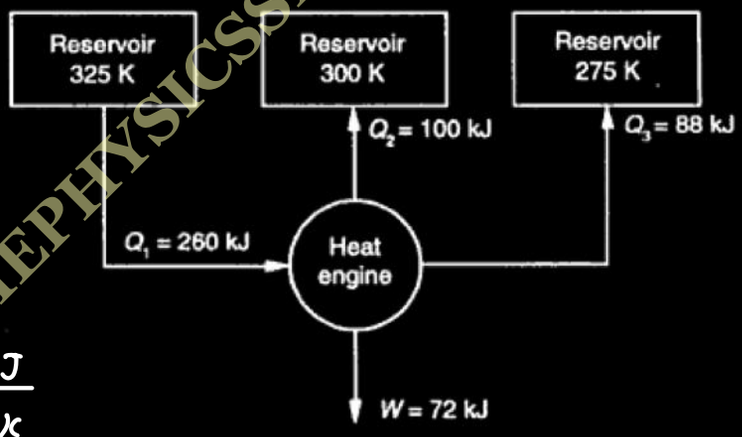
$\therefore \quad \oint \frac{dQ}{T} < 0$ for irreversible (Practical) Engine.

(∵ For Carnot: $\oint \frac{dQ}{T} = \frac{Q_H}{T_H} - \frac{Q_{L_R}}{T_L}$, while here $\oint \frac{dQ}{T} = \frac{Q_H}{T_H} - \frac{Q_{L_I}}{T_L}$

and $Q_{L_I} > Q_{L_R}$ (I → irreversible, R → Reversible).

EXAMPLE: An inventor claims to have designed a heat engine which absorbs 260 kJ of energy as heat from a reservoir at 52°C and delivers 72 kJ of work. He also states that the engine rejects 100 kJ and 88 kJ of energy to reservoirs at 27°C and 2°C respectively. Judge whether the claim is acceptable or not.

Solution: The schematic of the engine is as shown in the diagram. We shall check whether the engine satisfies clausius inequality.



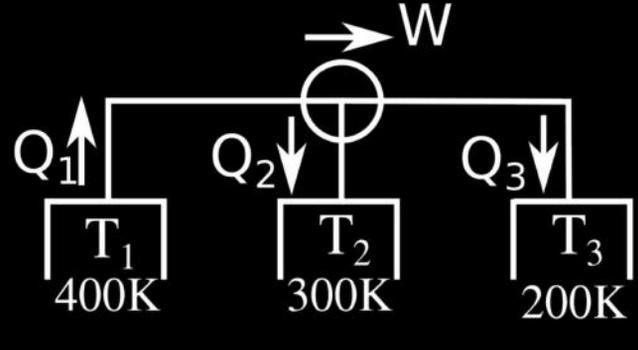
Now for our engine, $\oint \delta Q/T$

$$= \frac{260\,kJ}{325K} - \frac{100\,kJ}{300K} - \frac{88\,kJ}{275K}$$

$$\Rightarrow \oint \frac{\delta Q}{T} = +147\ J/K > 0$$

Thus it violates Clausius Inequality and ∴ it is an impossible Engine.

EXAMPLE: During a number of complete cycles a reversible engine (shown by a circle) absorbs 1200 J from reservoirs at 400 K and performs 200 J of mechanical work then:



(a) Reservoir at T_2 absorbs 200 J of heat while reservoir at T_3 loses 1200 J of heat

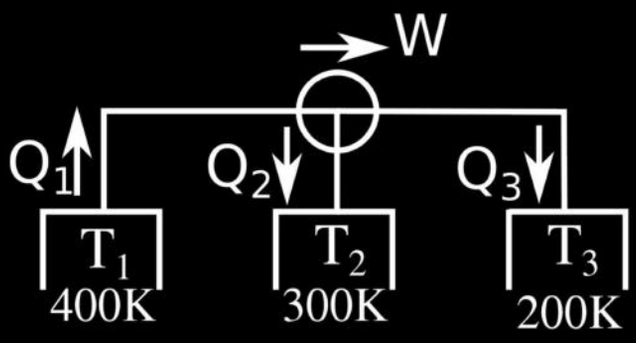
(b) The reservoir at T_2 absorbs 1200 J of Heat while that at T_3 loses 200 J of heat

(c) Change in entropy of reservoir at T_2 will be 4 J/K. in each cycle

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(d) Change in entropy of reservoir at T_3 will be $-1\ J/K$ in each cycle.

Solution: $Q_1 = 1200\ J$, $\Delta W = 200\ J$



$\Rightarrow Q_1 - Q_2 - Q_3 = 200$

or $1200 - Q_2 - Q_3 = 200$ ①

Also for reversible cyclic engines,

$\oint \frac{dQ}{T} = 0 \Rightarrow \frac{Q_1}{400} - \frac{Q_2}{300} - \frac{Q_3}{200} = 0$

$\Rightarrow \frac{1200}{400} - \frac{Q_2}{300} - \frac{Q_3}{200} = 0$ ②

From ① & ②, $Q_2 = 1200\ J$ $Q_3 = -200\ J$

$\Rightarrow$ Reservoir at T_2 gain 1200 J, That at T_3 loses 200 J (b)

Now ΔS of reservoir at $T_2 = \frac{Q_2}{T_2} = \frac{1200}{300} = 4\ J/K$.

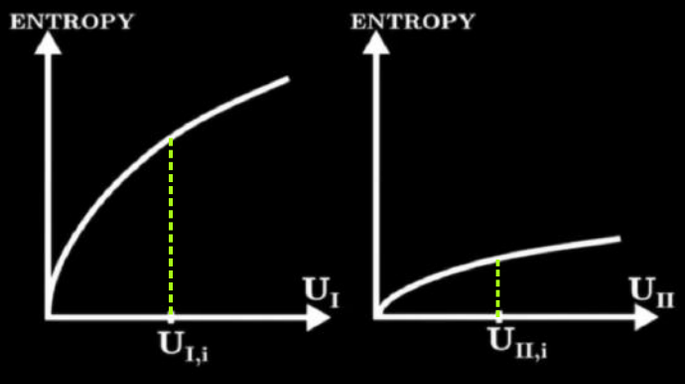
ΔS of reservoir at $T_3 = \frac{-200}{200} = -1\ J/K$

$\Rightarrow$ (C) and (d) are correct.

EXAMPLE: KVPY:

KVPY 2018 SX| ENTROPY CHALLENGE (Q96)

Graphs below show the entropy vs energy (U) of two systems I and II at constant volume. The initial energies of the systems are indicated by $U_{I,i}$ and $U_{II,i}$. respectively. Graphs are drawn to the same scale. The systems are then brought into thermal contact with each other. Assume that at all times the combined energy of the two systems remains constant. Choose the most appropriate option indicating the energies of the two systems and the total entropy after they achieve the equilibrium.



A. U_I increases and U_{II} decreases and the total entropy remains the same.
B. U_I decreases and U_{II} increases and the total entropy remains the same.
C. U_I increases and U_{II} decreases and the total entropy increases.
D. U_I decreases and U_{II} increases and the total entropy increases.

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Solution: At constant Volume, $dW = 0 \Rightarrow dQ = dU$.

Now $dS = \frac{dQ}{T} = \frac{dU}{T} \Rightarrow \frac{dS}{dU} = \frac{1}{T} \Rightarrow$

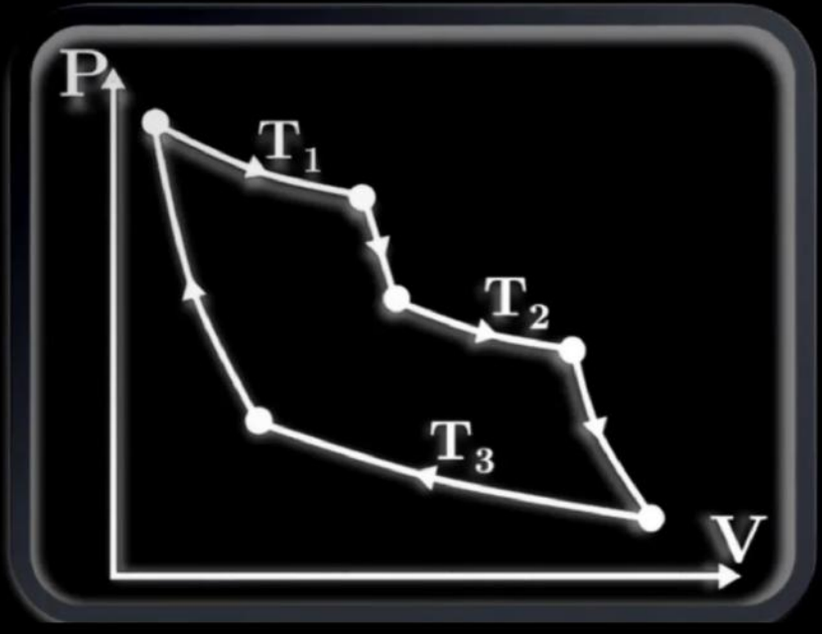
Slope of the S-U graph is inversely proportional to temperature
$\Rightarrow$ system II is at higher temperature $\because$ it has lower slope.
Now since energy flows from higher temperature to lower temperature
$\therefore U_{II}$ decreases and U_I increases. (Heat flows from II to I).
Also $\Delta S_{net} = -\frac{dQ}{T_{II}} + \frac{dQ}{T_I} > 0 \quad (\because T_I < T_{II})$

$\therefore$ Total entropy of the system increases.

Alternate Solution (Using Statistical Definition of Entropy): Since for same total energy, system I has higher entropy $\therefore$ it must offer more permutations for same total Energy and space $\therefore$ it must have more number of oscillators (Total degree of freedom). Now if same energy is distributed in more number of oscillators, the average energy per oscillator is lower and $\therefore$ system I has lower temperature. (After this we can argue similarly to previous method).

EXAMPLE: IRODOV

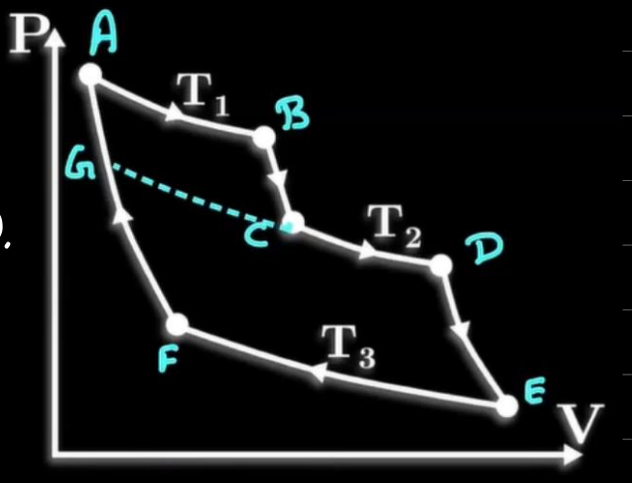
2.116. An ideal gas goes through a cycle consisting of alternate isothermal and adiabatic curves (Fig). The isothermal processes proceed at the temperatures T_1, T_2, and T_3. Find the efficiency of such a cycle, if in each isothermal expansion the gas volume increases in the same proportion.



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Solution: We visualise the cycle using two carnot cycles that we obtain by extending the isotherm DC.



According to Question, $\frac{V_B}{V_A} = \frac{V_D}{V_C} = r$ (say).

Also according to standard Carnot theory:

$\frac{V_C}{V_G} = \frac{V_B}{V_A} = r$, also, $\frac{V_E}{V_F} = \frac{V_D}{V_G}$

$\Rightarrow \frac{V_E}{V_F} = \frac{V_D}{V_C} \times \frac{V_C}{V_G} = r^2$

Now Recall that efficiency of a cycle = $\Sigma Q_W / \Sigma Q_H$

$\Rightarrow \eta = \frac{\Sigma Q_H - \Sigma |Q_L|}{\Sigma Q_H} = 1 - \frac{\Sigma |Q_L|}{\Sigma Q_H}$

In our process, heat is absorbed along isotherms A→B and C→D and Rejected along EF. Also for isothermal processes, $|Q| = nRT \ln(r)$ where r is the volume ratio (Higher/lower) Volume.

Here:

$Q_{H_{A\to B}} = nRT_1 \ln(r)$ $\quad Q_{H_{C\to D}} = nRT_2 \ln(r)$

$Q_{L_{E\to F}} = nRT_3 \ln(r^2) = 2nRT_3 \ln(r)$

$\Rightarrow \eta = 1 - \frac{\Sigma Q_L}{\Sigma Q_H} = 1 - \frac{2nRT_3 \ln(r)}{nRT_1 \ln(r) + nRT_2 \ln(r)}$

$\Rightarrow \eta = 1 - \left(\frac{2T_3}{T_1 + T_2}\right)$ Ans.

EXAMPLE: (IRODOV 2.138): An ideal gas with adiabatic exponent γ goes through a process $p = p_0 - \alpha V$ where p_0 and α are positive constants. At what volume will the gas have maximum entropy?

Solution: For ideal Gas, $dS = \frac{dQ}{T} \Rightarrow \frac{dS}{dV} = \frac{1}{T}\left(\frac{dQ}{dV}\right)$

For maximum entropy, $\frac{dS}{dV} = 0 \Rightarrow \frac{1}{T}\left(\frac{dQ}{dV}\right) = 0$

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$$\Rightarrow \frac{1}{T}\left[\frac{P\,dV + dU}{dV}\right] = 0 \Rightarrow P + \frac{dU}{dV} = 0 \quad (1)$$

Now $U = \frac{f}{2}PV = \frac{1}{\gamma-1}(PV) \Rightarrow \frac{dU}{dV} = \frac{1}{\gamma-1}\left(P + V\frac{dP}{dV}\right)$ (2)

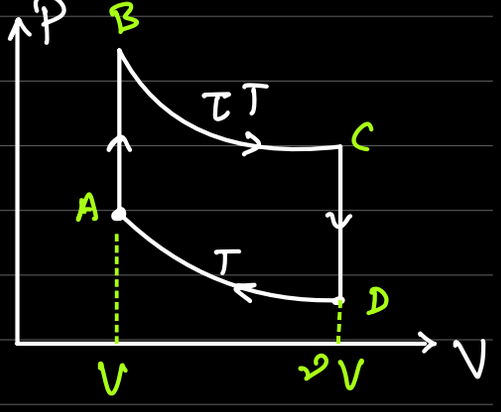
Now $P = P_0 - \alpha V \Rightarrow \frac{dP}{dV} = -\alpha \Rightarrow \frac{dU}{dV} = \frac{1}{\gamma-1}(P - \alpha V)$ (3)

Using (3) in (1), $P + \frac{1}{\gamma-1}(P - \alpha V) = 0$

$$\Rightarrow (P_0 - \alpha V) + \frac{1}{\gamma-1}(P_0 - 2\alpha V) = 0 \Rightarrow V = \frac{\gamma P_0}{\alpha(1+\gamma)} \quad \text{Ans.}$$

EXAMPLE: IRODOV 2.125: Find the efficiency of a cycle consisting of two isochoric and two isothermal lines if the volume varies γ fold and the absolute temperature varies τ fold within the cycle. γ of the gas is given.

Solution: Here Heat is rejected in C→D and D→A processes while it is absorbed in processes AB and BC.



Now $Q_{H_{AB}} = \frac{nR}{(\gamma-1)}[\tau - 1]T \quad [\because nC_v\Delta T]$.

$$Q_{H_{BC}} = nRT\tau \ln(\gamma)$$

$$|Q_{L_{CD}}| = \frac{nR}{\gamma-1}(\tau-1)T \quad (\because nC_v\Delta T)$$

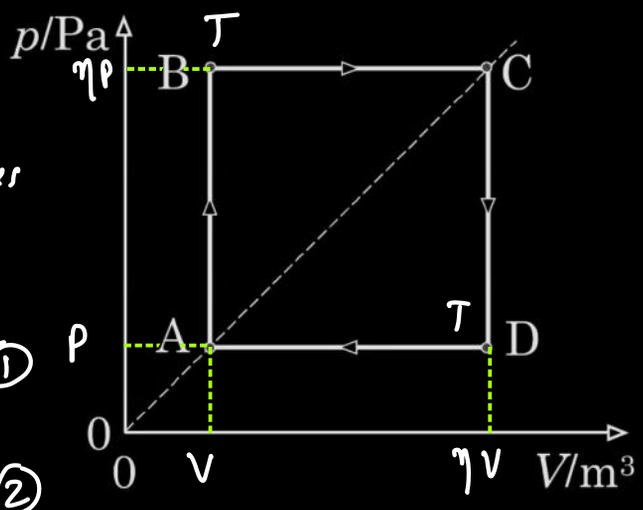
$$|Q_{L_{CD}}| = nRT\ln(\gamma) \quad \text{Now} \quad \Delta W = Q_H - |Q_L| = nRT\ln(\gamma)(\tau-1)$$

$$Q_H = nRT[\tau\ln(\gamma) + (\tau-1)/(\gamma-1)]$$

$$\Rightarrow \eta = \frac{\Delta W}{\Delta Q_H} = \frac{(\tau-1)\ln(\gamma)}{\tau\ln(\gamma) + (\tau-1)/(\gamma-1)} \quad \text{Ans.}$$

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EXAMPLE: In the shown graph, B and D lie on the same isotherm, Prove that Temperature at B is geometric mean of Temperatures at A and C



Solution: Between A & B: $\frac{PV}{T_A} = \frac{\eta PV}{T}$ ①

Similarly Between C & D: $\frac{\eta^2 PV}{T_C} = \frac{\eta PV}{T}$ ②

① × ② $\Rightarrow \frac{(\eta PV)^2}{T_A T_C} = \frac{(\eta PV)^2}{T^2} \Rightarrow T = \sqrt{T_A T_C}$ Q.E.D.

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